

Experiment No: 4

AIM: To develop a blockchain-powered web application using Solidity programming language on Remix IDE and embedding Metamask Wallet

THEORY:

Blockchain is a **distributed and tamper-resistant digital ledger** used to record transactions and data. In this experiment, blockchain technology is used to implement a **Digital Warranty and Product Ownership Transfer System**.

The system uses a **Solidity smart contract** deployed on the **Ethereum Sepolia Testnet** through **Remix IDE**. MetaMask acts as the blockchain wallet and is used to connect the user's account with Remix and authorize transactions.

The smart contract stores product information such as:

- Product ID
- Product name
- Owner's blockchain address
- Warranty period
- Purchase date
- Product existence status

The contract also provides functions to register a product, check its warranty status, retrieve product information, and transfer ownership.

A major blockchain feature demonstrated in this project is **access control through the smart contract**. The `transferOwnership()` function contains a condition that allows only the current owner to transfer the product:

```
require(  
    products[_productId].owner == msg.sender,  
    "Only current owner can transfer"  
);
```

Therefore, when Account 1 attempted to transfer the product after ownership had already been transferred to Account 2, the transaction was rejected with:

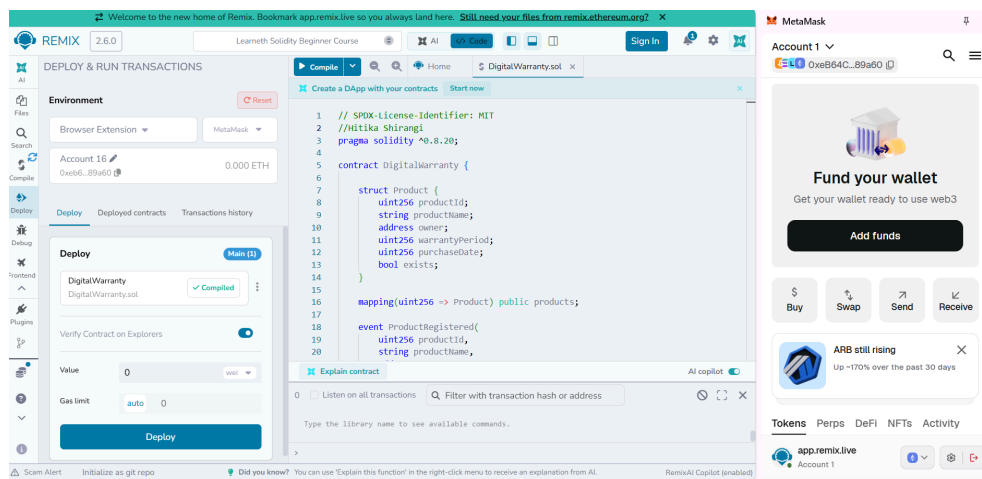
Only current owner can transfer

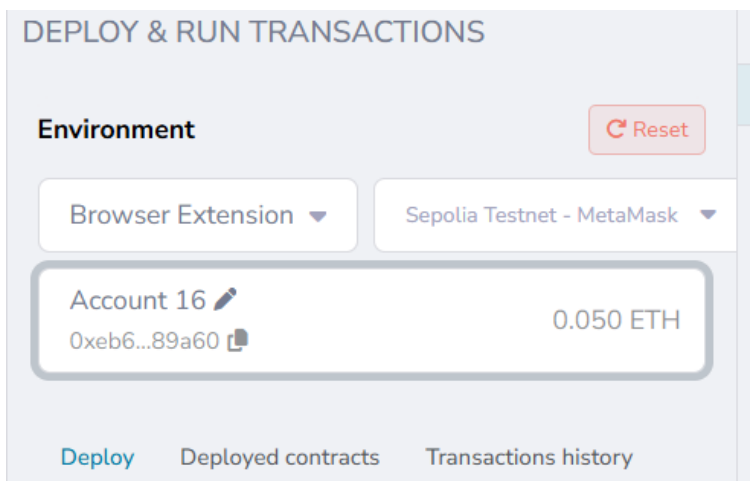
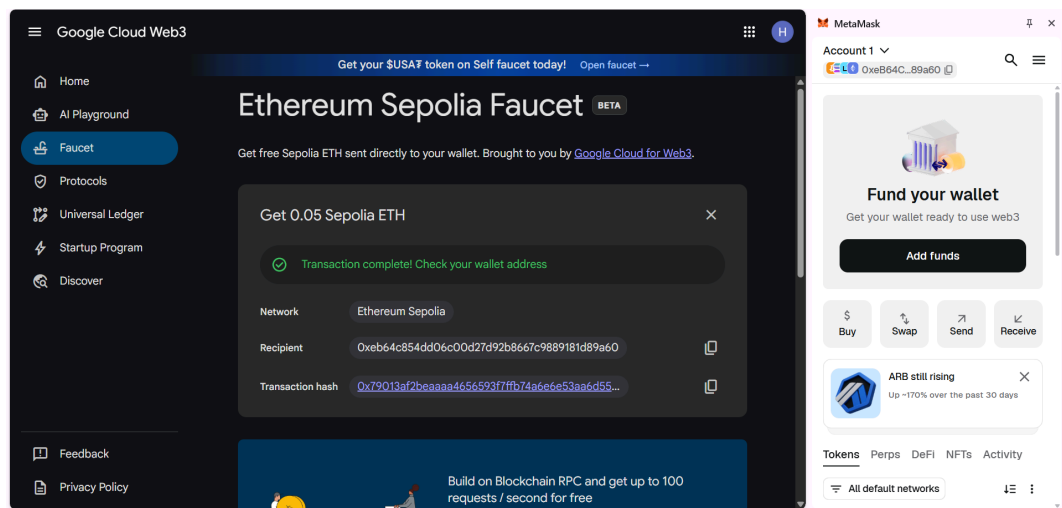
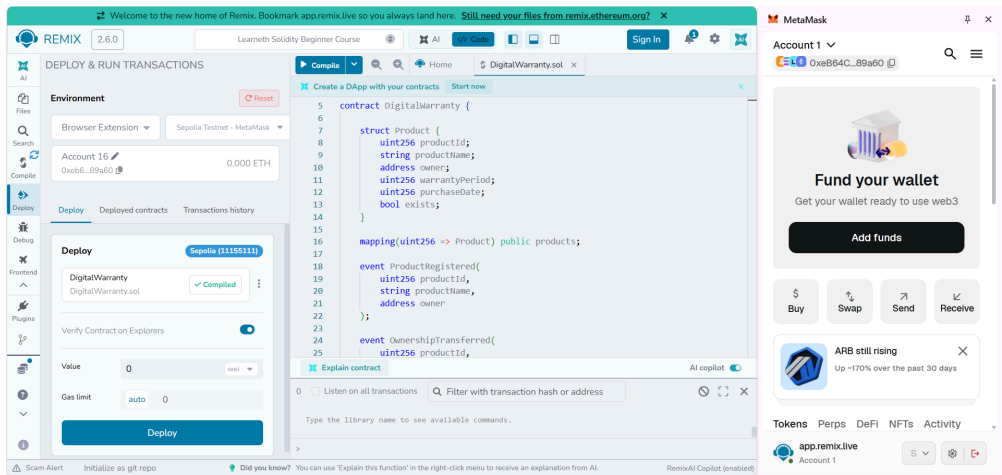
This demonstrates that the smart contract automatically enforces the ownership rule.

The experiment follows the lab requirements of setting up MetaMask, connecting a testnet to Remix, creating and deploying a Solidity smart contract, checking transaction details, and interacting with the deployed contract.

STEPS and OUTPUT:

1. Setup metamask and connect with Remix IDE
2. Create the solidity smart contract
3. Compile and deploy it.
4. Register a product
5. Verify product details
6. Check warranty
7. Transfer product ownership
8. Verify new ownership
9. Demonstrate unauthorized transfer restriction





REMIX2.6.0

Learneth Solidity Beginner Course

AI

VS Code

Home

Sign In

DEPLOY & RUN TRANSACTIONS

Environment

Browser Extension

Sepolia Testnet - MetaMask

Account 16

0.050 ETH

Deploy

Deployed contracts

Transactions history

Deploy

Sepolia (11155111)

DigitalWarranty

DigitalWarranty.sol

Verify Contract on Explorers

Value

0

Gas limit

auto

0

Deploy

contract DigitalWarranty {

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Functions

● **checkWarranty** (uint256) Call

Calldata  Params 

● **getProduct** (uint256) Call

Calldata  Params 

● **products** (uint256) Call

Calldata  Params 

● **registerProduct** (uint256, string, uint256)

Calldata  Params 

Transact

● **transferOwnership** (uint256, address)

Calldata  Params 

Transact

DEPLOY & RUN TRANSACTIONS

Environment

Browser Extension

Sepolia Testnet - MetaMask

Account 16

0xeb6...89a60

0.047 ETH

Deploy

Deployed contracts

Transactions history

products (uint256)

param0 (uint256)

Calldata Params

registerProduct (uint256, string, uint256)

1001

Samsung Refrigerator

365

Calldata Params

Transact

transferOwnership (uint256, address)

_productId (uint256)

_newOwner (address)

Compile

Create a DApp with your contracts

Start now

```
5 contract DigitalWarranty {
6
7     struct Product {
8         uint256 productId;
9         string productName;
10        address owner;
11        uint256 warrantyPeriod;
12        uint256 purchaseDate;
13        bool exists;
14    }
15
16    mapping(uint256 => Product)
17
18    event ProductRegistered(
19        uint256 productId,
20        string productName,
21        address owner
22    );
23
24    event OwnershipTransferred(
25        uint256 productId,
```

Explain contract

AI copilot

0

Listen on all transactions

AI USE STRINGS/NUMBERS FOLLOW THE RULES

Account 1

Transaction request

Balance changes

No changes

Added protection

Optional

Because you're interacting with an unknown address, protection can prevent some malicious transactions.

Learn more

Network

Sepolia

Request from

app.remix.live

Interacting with

0x6BCC5...7a52E

Network fee

0.0008 SepoliaETH

Speed

Market ~12 sec

Cancel

Confirm

getProduct (uint256)

Call

1001

Calldata Params

0:uint256: 1001

1:string: Samsung Refrigerator

2:address: 0xB64C854dD06c00D27d92B8667C9889181d89a60

3:uint256: 365

4:uint256: 1789743204

5:bool: true

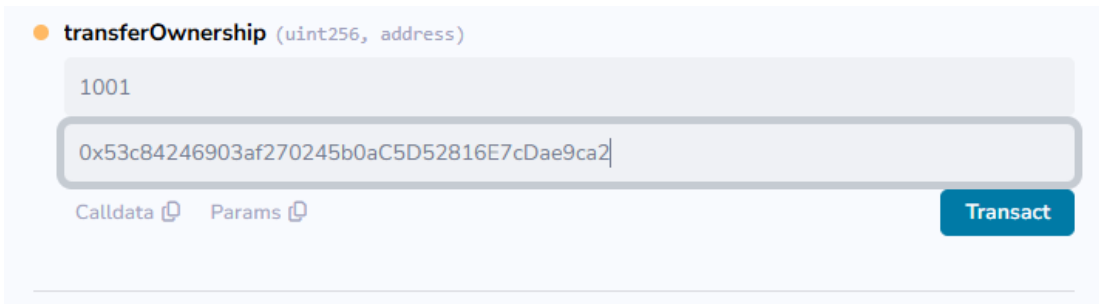
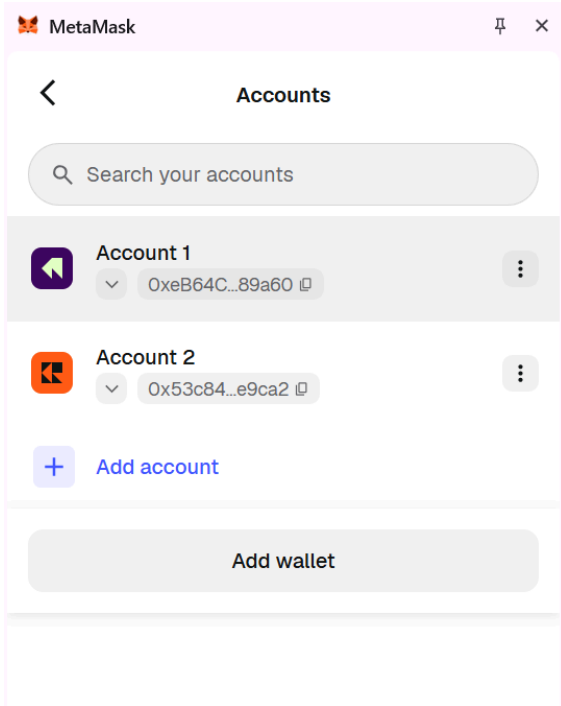
checkWarranty (uint256)

Call

1001

Calldata Params

0:bool: true





Account 1



Transaction request

Balance changes ⓘ

No changes

ⓘ Added protection ⓘ

Optional



Because you're interacting with an unknown address, protection can prevent some malicious transactions.

[Learn more](#)

Network

Ⓢ Sepolia

Request from ⓘ

app.remix.live

Interacting with ⓘ

🟢 0x6BCC5...7a52E

Network fee ⓘ

✎ 0.0004 Ⓢ SepoliaETH

Speed

Market ~12 sec

Cancel

Confirm

● **getProduct** (uint256)

Call

1001

Calldata ⓘ Params ⓘ

0:uint256: 1001

1:string: Samsung Refrigerator

2:address: 0x53c84246903af270245b0aC5D52816E7cDae9ca2

3:uint256: 365

4:uint256: 1789743204

5:bool: true

